OBSCYRATECH

CYBERSECURITY • TRAINING • SERVICES

PROFESSIONAL SYLLABUS

Fundamentals

A complete beginner-friendly foundation covering security principles, threats, networks, identity, cryptography, incident response, governance, and emerging trends — your launchpad for Ethical Hacking, VAPT, SOC, Cloud Security, and GRC pathways.

MODULES	LEVEL	FORMAT	OUTCOME
15 Modules	**Beginner Friendly**	**Theory + Projects**	**Foundation Ready**

obscyratech@gmail.com

+91 96062 62858 | +91 91104 60406

Mysore, Karnataka, India

Course Description

Cybersecurity Fundamentals provides students with a strong foundation in protecting digital systems, networks, applications, and information from cyber threats. The course introduces essential cybersecurity concepts, common attacks, security controls, risk management, and security best practices.

The curriculum is designed for **beginners** and focuses on practical understanding, security awareness, and foundational skills needed before progressing into specialized domains such as **Ethical Hacking, Network Security, Cloud Security, Digital Forensics,** or **Security Operations**.

Syllabus Index

A complete pathway from cybersecurity basics to capstone — 15 structured modules building the perfect industry foundation.

CORE CURRICULUM

Course Modules

MODULE 01 Introduction to Cybersecurity

LEARNING OBJECTIVES

- Understand the role of cybersecurity in modern organizations.
- Learn basic cybersecurity terminology.
- Explore career opportunities in cybersecurity.

KEY TOPICS

- What is Cybersecurity?
- Importance of Information Security
- Cybersecurity Domains
- Threat Landscape Overview
- Cybersecurity Careers

MINI PROJECT Analyze a recent cyberattack and discuss its impact.

MODULE 02 Information Security Principles

LEARNING OBJECTIVES

- Understand the foundations of information security.
- Apply security principles to real-world situations.
- Identify security risks.

KEY TOPICS

- CIA Triad (Confidentiality, Integrity, Availability)
- Risk, Threat, and Vulnerability
- Security Controls
- Defense in Depth
- Security Framework Concepts

MINI PROJECT Create a security plan for protecting personal data.

MODULE 03 Computer Systems & Operating System Security

LEARNING OBJECTIVES

- Understand how computer systems operate.
- Learn basic system security concepts.
- Recognize operating system vulnerabilities.

KEY TOPICS

- Hardware and Software Basics
- Windows Fundamentals
- Linux Fundamentals
- User Accounts and Permissions
- System Security Concepts

MINI PROJECT Perform a basic security review of a computer system.

MODULE 04 Networking Fundamentals for Cybersecurity

LEARNING OBJECTIVES

- Understand how networks function.
- Learn essential networking terminology.
- Recognize network communication processes.

KEY TOPICS

- Network Basics
- TCP/IP Model
- IP Addressing
- Common Network Devices
- Internet Communication

MINI PROJECT Create a network diagram for a small business.

MODULE 05 Network Security Essentials

LEARNING OBJECTIVES

- Understand how networks are protected.
- Learn about security technologies.
- Identify common network attacks.

KEY TOPICS

- Firewalls
- Intrusion Detection Systems
- Network Segmentation
- Wireless Security
- Secure Communication

MINI PROJECT Design a secure network architecture.

MODULE 06 Cyber Threats & Malware

LEARNING OBJECTIVES

- Identify common cyber threats.
- Understand malware behavior.
- Learn basic threat mitigation techniques.

KEY TOPICS

- Viruses
- Worms
- Trojans
- Ransomware
- Spyware and Adware

MINI PROJECT Research a famous malware attack and present findings.

MODULE 07 Social Engineering & Human Security

LEARNING OBJECTIVES

- Recognize human-based cyber attacks.
- Understand attacker psychology.
- Develop security awareness skills.

KEY TOPICS

- Social Engineering Fundamentals
- Phishing
- Spear Phishing
- Business Email Compromise
- Security Awareness

MINI PROJECT Analyze phishing emails and identify indicators.

MODULE 08 Identity & Access Management

LEARNING OBJECTIVES

- Understand authentication concepts.
- Learn access control methods.
- Improve account security practices.

KEY TOPICS

- Authentication
- Authorization
- Password Security
- Multi-Factor Authentication (MFA)
- Access Control Models

MINI PROJECT Create a secure password and authentication policy.

MODULE 09 Cryptography Fundamentals

LEARNING OBJECTIVES

- Understand encryption concepts.
- Learn how cryptography protects data.
- Recognize secure communication methods.

KEY TOPICS

- Encryption Basics
- Symmetric Encryption
- Asymmetric Encryption
- Hashing
- Digital Signatures

MINI PROJECT Map where encryption is used in daily life.

MODULE 10 Vulnerability Management & Security Assessment

LEARNING OBJECTIVES

- Understand security weaknesses.
- Learn vulnerability management processes.
- Recognize the importance of updates and patching.

KEY TOPICS

- Vulnerabilities
- Risk Assessment
- Security Audits
- Patch Management
- Security Assessments

MINI PROJECT Conduct a vulnerability assessment checklist.

MODULE 11 Incident Response & Business Continuity

LEARNING OBJECTIVES

- Understand how organizations respond to incidents.
- Learn disaster recovery concepts.
- Develop incident handling awareness.

KEY TOPICS

- Incident Response Lifecycle
- Security Monitoring
- Disaster Recovery
- Business Continuity
- Crisis Management

MINI PROJECT Create an incident response plan for a small company.

MODULE 12 Cybersecurity Governance, Ethics & Compliance

LEARNING OBJECTIVES

- Understand legal and ethical responsibilities.
- Learn governance and compliance concepts.
- Recognize privacy requirements.

KEY TOPICS

- Cybersecurity Ethics
- Cybercrime Laws
- Privacy and Data Protection
- Governance Frameworks
- Security Policies

MINI PROJECT Draft a basic acceptable-use policy.

MODULE 13 Emerging Technologies & Security Trends

LEARNING OBJECTIVES

- Explore modern cybersecurity challenges.
- Understand security implications of new technologies.
- Stay aware of industry trends.

KEY TOPICS

- Cloud Security Fundamentals
- Internet of Things (IoT) Security
- Artificial Intelligence Security
- Mobile Security Basics
- Future Threat Landscape

MINI PROJECT Research an emerging technology and its security risks.

MODULE 14 Security Awareness & Best Practices

LEARNING OBJECTIVES

- Apply cybersecurity principles in daily life.
- Develop safe online habits.
- Promote security culture.

KEY TOPICS

- Personal Cyber Hygiene
- Secure Browsing
- Safe Use of Social Media
- Data Protection Practices
- Security Awareness Programs

MINI PROJECT Create a cybersecurity awareness campaign.

MODULE 15 Capstone Project & Career Roadmap

LEARNING OBJECTIVES

- Integrate knowledge from all modules.
- Demonstrate cybersecurity understanding.
- Plan future learning and career development.

KEY TOPICS

- Security Assessment Methodology
- Risk Identification
- Security Recommendations
- Cybersecurity Career Paths
- Certification Roadmap

FINAL CAPSTONE PROJECT

Perform a cybersecurity assessment of a fictional organization covering:

- ✓ Risks and Threats
- ✓ Network Security
- ✓ Access Controls
- ✓ Security Policies
- ✓ Incident Response Recommendations

Recommended Beginner Tools

Industry-standard, beginner-friendly tools used throughout the practical activities:

- **Wireshark**
  Capture & analyze live network traffic and packets.
- **VirtualBox**
  Create isolated virtual machines for safe security labs.
- **Cisco Packet Tracer**
  Visualize and simulate networks & security devices.
- **Nmap**
  Beginner-friendly network scanning & service discovery.

Assessment & Grading

A balanced mix of assignments, quizzes, practical activities, midterm and a final capstone project.

COMPONENT	WEIGHTAGE
Module Assignments	25%
Quizzes	15%
Practical Activities	20%
Midterm Assessment	15%
Final Capstone Project	25%
Total	**100%**

Course Outcome & Next Steps

After completing this course, students will understand the fundamental concepts of cybersecurity and be prepared to pursue **specialized cybersecurity courses** across multiple career domains. This makes it an ideal **foundation course for any cybersecurity learning pathway**.

Recommended advanced tracks at Obscyratech:

- **Ethical Hacking**
- **Network Security**
- **Cloud Security**
- **Digital Forensics**
- **Security Operations Center (SOC)**
- **Governance, Risk & Compliance (GRC)**
- **Vulnerability Assessment & Penetration Testing (VAPT)**

Program Promise: Cybersecurity Fundamentals is the **perfect launchpad** for absolute beginners — students, IT professionals, and career switchers — to enter the cybersecurity industry with confidence. Every module blends concepts, mini projects, and real-world scenarios to prepare you for advanced certifications and specialized careers.

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Lay the perfect — for every career path ahead.

Join Obscyratech's Cybersecurity Fundamentals program and build the knowledge, mindset, and skills to launch into any advanced cybersecurity track — Ethical Hacking, VAPT, SOC, Cloud Security, GRC, and beyond.

GET IN TOUCH

EMAIL

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PHONE

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LOCATION

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PROGRAM

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